We know the number of symbols emitted is the length of the protein

When you have deletions, you can't make them their own columns
Instead, within each column you have deletion and insertions

In proteins, a single mutation can cause large structural changes
yet you can change many amino acids and they fold the same

Evolutionary Trees

The Black Death was slow to spread

Trees:
- Connected - Life originates from one point
- Acyclic - Species are collections of organisms that produce viable offspring

Leaves: Present-day species   Internal-nodes: Ancestors

Dinosaurs divide out into two groups

Saurischia
Ornithischia   (based on hip bones)

Dollo's principle of irreversibility (1893): Evolution doesn't re-invent the same organ

However, this is not right

Distance Matrix

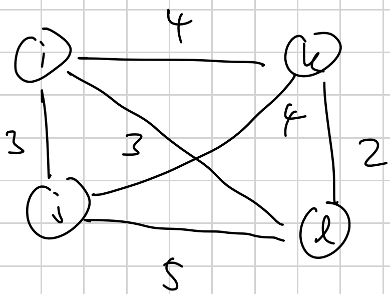
$D_{i,j} = D_{j,i}$
$D_{i,j} \geq 0$ (Metric Space)
$D_{i,k} \leq D_{i,j} + D_{j,k}$

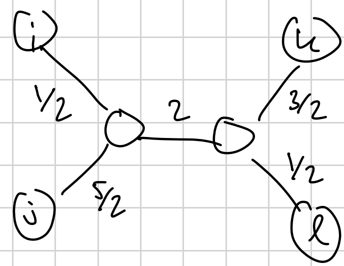
Distance-Based Phylogeny Problem

Input: Distance Matrix
Output: A tree fitting this Distance Matrix

Construct a tree s.t. the distance between two leaves is exactly the same as in the Distance Matrix

$$\begin{matrix} 0 & 3 & 4 & 3 \\ 3 & 0 & 4 & 5 \\ 4 & 4 & 0 & 2 \\ 3 & 5 & 2 & 0 \end{matrix}$$





$a + b = 3$
$a + c + e = 4$
$a + c + d = 3$
$b + c = 9/2$
$a + c = 5/2$
$a + b = 3$

$2a + 2b + 2c = 10$
$a + b + c = 5$

$b = c + d$
$e - d = 1$
$e + d = 2$

$e = 3/2$
$d = 1/2$

Sometimes No Tree Fits!

Matrix is Additive if
$D_{i,j} + D_{k,l}$ $D_{i,k} + D_{j,l}$ $D_{i,l} + D_{j,k}$

2 must be equal, third is less
Must hold for all quartets

Simple Tree: No nodes of degree 2
Thm) Additive Matrices have a unique simple tree fitting

Pick 2 Nodes
Merge them, figure out distances with algebra
Make a new node representing them => recurse